

Adiposity and insulin resistance in humans: the role of the different tissue and cellular lipid depots.



Human adiposity has long been associated with insulin resistance and increased cardiovascular risk, and abdominal adiposity is considered particularly adverse. Intra-abdominal fat is associated with insulin resistance, possibly mediated by greater lipolytic activity, lower adiponectin levels, resistance to leptin, and increased inflammatory cytokines, although the latter contribution is less clear. Liver lipid is also closely associated with, and likely to be an important contributor to, insulin resistance, but it may also be in part the consequence of the lipogenic pathway of insulin action being up-regulated by hyperinsulinemia and unimpaired signaling. Again, intramyocellular triglyceride is associated with muscle insulin resistance, but anomalies include higher intramyocellular triglyceride in insulin-sensitive athletes and women (vs men). Such issues could be explained if the "culprits" were active lipid moieties such as diacylglycerol and ceramide species, dependent more on lipid metabolism and partitioning than triglyceride amount. Subcutaneous fat, especially gluteofemoral, appears metabolically protective, illustrated by insulin resistance and dyslipidemia in patients with lipodystrophy. However, some studies suggest that deep sc abdominal fat may have adverse properties. Pericardial and perivascular fat relate to atheromatous disease, but not clearly to insulin resistance. There has been recent interest in recognizable brown adipose tissue in adult humans and its possible augmentation by a hormone, irisin, from exercising muscle. Brown adipose tissue is metabolically active, oxidizes fatty acids, and generates heat but, because of its small and variable quantities, its metabolic importance in humans under usual living conditions is still unclear. Further understanding of specific roles of different lipid depots may help new approaches to control obesity and its metabolic sequelae.